

Modeling Human-AI Collaboration Dynamics: A Knowledge Graph Approach to Autonomy, Explainability, and Outcome Analysis

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Abstract

Human-AI collaboration is increasingly applied across domains such as healthcare, scientific research, and education, where decisions have significant consequences. Understanding how key factors such as AI autonomy and explainability influence human agreement and overall outcomes is critical for designing effective and trustworthy systems [5, 4]. In this paper, we investigate the role of these factors within Human-AI collaborative settings using Knowledge Graph engineering methods.

We construct a knowledge graph consisting of 626 entities, including 21 use cases derived from a curated set of 7 research papers in the Human-Centered AI (HHAI) domain [1, 2, 3, 4, 5, 6, 7], selected to represent diverse application areas. In addition, we model six Human-AI collaboration scenarios. Three scenarios are taken from existing HI competition settings: Scenario 1 (T3 – HICentreCM 2024, Healthcare Diagnosis Team “Dr. House”, Worst Case), Scenario 2 (T1 – HHAI 2024, Research HI Team “Dream Team”, Best Case), and Scenario 3 (T3 – HHAI 2024, Healthcare Diagnosis Team “#TeamSusan”, Best Case). To complement these extremes, three additional scenarios were manually designed to represent intermediate outcomes, capturing collaboration situations that are neither strongly successful nor clearly failing. These scenarios are modeled as workflows of interactions, decisions, and outcomes, enabling structured analysis of Human-AI collaboration dynamics. Through symbolic querying and analysis, we examine how varying levels of autonomy and explainability influence human agreement and decision quality.

Our findings suggest that high AI autonomy can lead to improved outcomes when paired with strong explainability, while low explainability—regardless of autonomy—can result in misinterpretation, reduced trust, and poor outcomes. Furthermore, partial explainability and misaligned human agreement often lead to inconsistent or mixed results.

These results highlight the importance of balancing autonomy and transparency in Human-AI systems and demonstrate the value of Knowledge Graphs for modeling and analyzing complex collaborative behaviors. The study provides insights into designing more reliable and aligned Human-AI interactions across domains.

1 Introduction

The rapid integration of AI into high stakes decision environments has fundamentally reshaped professional practice across healthcare, scientific research, and education. Rather than displacing human judgment, contemporary AI systems increasingly function as collaborative partners, a mode of operation collectively theorized under the Hybrid Intelligence (HI) paradigm [3, 2]. In HI settings, collaboration quality depends not only on the technical capabilities of the AI component but on the structure of the interaction itself: how much autonomy the AI exercises, how transparently it communicates its reasoning, and how well human and AI judgments converge over time.

Despite growing recognition that these dimensions matter, the research community still lacks structured, queryable representations of how autonomy, explainability, and agreement jointly shape col-

laborative outcomes. Most existing analyses treat these factors in isolation, making it difficult to reason systematically across scenarios or identify configurations reliably associated with good or poor performance. Knowledge Graphs offer a principled response to this gap by encoding scenarios as structured instances over a shared ontology, they support symbolic querying, graph level analysis, and sub-symbolic generalization through embedding methods over learned relational patterns.

This paper presents a KG based investigation of Human-AI collaboration dynamics. We extend the Hybrid Intelligence Ontology (HI Ontology v2) with four analytical dimensions—autonomy level, explainability level, agreement level, and outcome quality—and instantiate six representative collaboration scenarios drawn from clinical decision support and neuro degenerative disease research. These scenarios span a deliberate range, from a best case configuration characterized by strong alignment, high explainability, and positive outcomes, to a worst case setting in which an opaque, highly autonomous AI produces poor outcomes through forced agreement. Together they constitute a small but structured laboratory for examining how these dimensions interact.

We analyze the resulting knowledge graph through three methods. First, SPARQL queries expose structural relationships between the four collaboration dimensions and outcome quality. Second, graph theoretic metrics degree distribution, clustering coefficient, betweenness centrality, and PageRank—characterize the topology of the full HI knowledge graph. Third, a ComplEx embedding model trained using the PyKEEN framework evaluates the feasibility of sub symbolic link prediction over the graph; its limitations are examined critically for what they reveal about the need for constraint aware approaches.

2 Data

2.1 Ontology Design

To support the analysis of Human-AI collaboration dynamics, we extended the original Hybrid Intelligence ontology with a focused and minimal set of additions. The goal of the extension was to explicitly model key factors that influence the quality of collaboration between human and AI agents, while keeping the ontology simple and queryable.

The main extension introduces four central dimensions that characterize interaction quality: *autonomy level*, *explainability level*, *agreement level*, and *outcome quality*. These were modeled as sub-classes of **Property Value**, enabling the representation of qualitative attributes of interactions in a structured way. Each dimension is instantiated through controlled individuals such as **HighAutonomy**, **LowExplainability**, **AlignedAgreement**, and **GoodOutcome**. This design allows consistent annotation and comparison across different scenarios.

To connect these dimensions to actual interactions, we introduced dedicated object properties, including **hasAutonomyLevel**, **hasExplainabilityLevel**, **hasAgreementLevel**, and **hasOutcomeQuality**. All of these properties have **Interaction** as their domain and the corresponding value class as their range. This ensures a clear and uniform structure for describing each collaboration instance.

In addition, we incorporated a lightweight process oriented layer to capture the core decision making flow in Human-AI interaction. This includes the classes **Recommendation**, **Explanation**, and **Decision**, which model how AI systems generate suggestions, provide reasoning, and how human agents ultimately make decisions. Corresponding properties such as **generatesRecommendation**, **providesExplanation**, and **makesDecision** were added to represent these relationships.

As an example, a scenario instance (e.g., **S3_RobotTeaching**) is modeled as an **Interaction** and linked to its corresponding properties as follows: it has **HighAutonomy**, **HighExplainability**, **AlignedAgreement**, and **GoodOutcome**. This structured representation enables systematic comparison of scenarios and supports querying patterns across different configurations of Human-AI collaboration.

Overall, the design prioritizes clarity, minimalism, and analytical usefulness, ensuring that the ontology remains easy to extend while supporting meaningful insights through querying and analysis.

2.1.1 Logical Constraints and Property Characteristics

To make the ontology more logically consistent and easier for automated reasoning, several OWL features were added.

Disjointness axioms were used to keep incompatible classes separate. For example, an entity cannot be both an **ArtificialAgent** and a **HumanAgent** at the same time. Similarly, the outcome classes

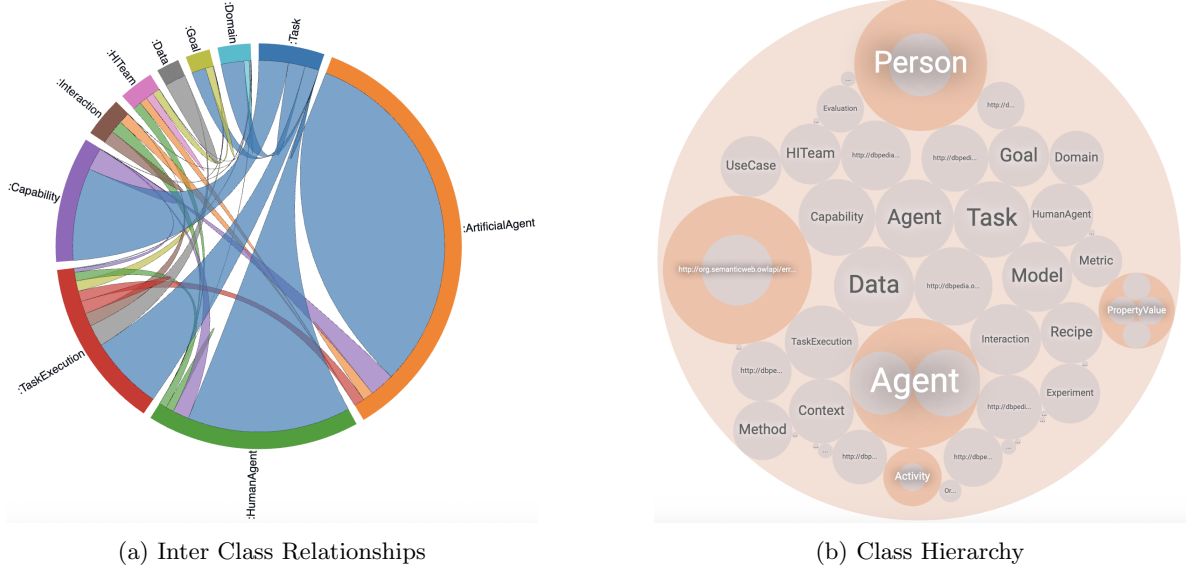


Figure 1: Ontology Structure visualisation

GoodOutcome, MixedOutcome, and BadOutcome are disjoint, so an interaction can only have one type of outcome.

Union classes make it easier to query and reason over multiple related agent types together. For example, CollaborativeAgent is defined as the union of HumanAgent and ArtificialAgent. This means queries using properties like hasCapability or involvedInInteraction can retrieve both human and AI agents without needing separate patterns for each type.

Functional properties ensure that some relationships can only have one value. For instance, hasOutcomeQuality is a functional property, meaning each interaction can have only one final outcome.

Inverse properties make it easier to navigate the knowledge graph in both directions. For example, allowsTask is the inverse of requiresCapability. This means you can find tasks from capabilities or capabilities from tasks without storing the same information twice.

Property chains capture more complex relationships across sequences of actions. For example, if an agent performsExecution of an activity that realizesTask, it can be inferred that the agent also executesTask. This allows reasoning engines to automatically infer relationships without explicitly stating them.

In addition to these constraints, existential and universal restrictions were introduced using owl:someValuesFrom and owl:allValuesFrom. These restrictions ensure that interactions and teams are semantically well-formed.

2.1.2 Competency Questions (CQs)

To guide the development of the HI-Ontology extension, we defined a set of Competency Questions (CQs) that the graph must be able to resolve. These CQs serve as the requirements for our data modeling and SPARQL analysis:-

- CQ1: Which combinations of AI autonomy and explainability levels are most frequently associated with a positive interaction outcome?
- CQ2: What is the standard team composition (ratio of human to artificial agents) across different domain use cases?
- CQ3: In scenarios where human-AI agreement is strictly "Forced," what are the resulting outcome qualities?
- CQ4: Which specific tasks require capabilities that are currently only provided by Artificial Agents?

2.2 Instance Creation

The knowledge graph contains 626 named individuals, including 21 Use Case instances derived from Human-Centered AI research papers and six representative Human-AI collaboration scenarios.

2.2.1 Extraction Process

The 21 use cases were derived from a curated set of 7 research papers in the Human-Centered AI (HHAI) domain [1, 2, 3, 4, 5, 6, 7]. through a semi structured extraction process supported by a large language model (LLM). The LLM was used to assist in identifying key elements of Human-AI collaboration described in the papers [1, 3]. For each paper, the extraction focused on three core components:

- The agents involved (e.g., human experts, AI systems, or hybrid teams)
- The Tasks performed
- The Goals pursued within the collaboration scenario.

The candidate entities and relationships suggested by the LLM were reviewed, validated, and integrated manually into the ontology during the knowledge graph construction process.

2.2.2 Structural Representation

Each use case is modeled as an instance of the class UseCase. These instances are structurally linked to:

- a HITeam representing the human-AI collective,
- a Domain (such as healthcare, education, or scientific research)
- a Goal describing the intended outcome of the collaboration.

This structure allows use cases to be compared across domains and enables queries that analyze how different team compositions or objectives influence collaboration outcomes.

2.2.3 Interaction Modeling

To provide deeper analytical insight, six representative scenarios were modeled in detail by decomposing them into Interactions. Each interaction is annotated with the four analytical dimensions introduced in the ontology extension - autonomy level, explainability level, agreement level and outcome quality.

To ensure a more realistic and process-oriented representation, interactions are not directly linked to use cases, but instead connected through intermediate TaskExecution instances. Each execution represents a concrete realization of a task within a specific use case and is linked to one or more interaction episodes via the property hasInteractionEpisode.

For example, clinical scenarios (S1 and S2) are linked through a shared execution instance (BrainAgeAssessmentExecution), while research scenarios (S5 and S6) are connected via a dedicated execution instance (ProteinResearchScenarioExecution). This design reflects the fact that interactions occur as part of task executions rather than existing independently.

This representation allows the knowledge graph to capture both high-level use cases and fine-grained interaction dynamics, enabling structured comparisons across collaboration configurations.

2.3 Linking to External Vocabularies

To improve interoperability and situate the knowledge graph within the broader Semantic Web ecosystem, we linked our ontology and instance data to three established vocabularies and one external knowledge base.

At the schema level, the `hi:Agent` class is declared equivalent to `prov:Agent` from the PROV-O provenance ontology [15], which is imported directly via `owl:imports`. The `hi:HumanAgent` class is declared as a subclass of `foaf:Person` [16], making human participants in collaboration scenarios compatible with FOAF-based person descriptions. Domain value classes (`AutonomyLevel`, `ExplainabilityLevel`,

AgreementLevel, OutcomeQuality) are anchored to schema:PropertyValue from Schema.org[17], enabling structured property-value descriptions compatible with that vocabulary.

At the instance level, the knowledge graph is enriched with links to both DBpedia and Wikidata[18, 19]. A total of over 117 individuals are connected to DBpedia entities using owl:sameAs.

To further enhance coverage and semantic grounding, 20 additional links to Wikidata entities were introduced. Wikidata complements DBpedia by providing structured, up-to-date, and highly curated entity representations, thereby improving the robustness and interoperability of the knowledge graph.

These integrations ensure that the ontology is not isolated but connected to the wider semantic web ecosystem, supporting reuse, extensibility, and potential federated querying against external SPARQL endpoint.

2.4 Ontology Expansion and Reasoning

The original merged ontology contained 1695 triples. During the development of the HI-Ontology extension, additional domain-specific classes, properties, and instances were manually introduced to model Human–AI collaboration scenarios in greater detail.

After incorporating these extensions and executing an OWL reasoner (HermiT), the final knowledge graph contained 3004 triples. The increase in graph size therefore reflects both manual ontology enrichment and automatically inferred triples generated through reasoning.

The reasoning process derives additional relationships implied by the ontology’s logical axioms, such as property hierarchies, inverse relations, and property chains. These inferred triples enrich the knowledge graph with implicit knowledge that is not explicitly asserted in the dataset.

Overall, the final graph represents an approximate 77% increase in total statements compared to the original merged ontology, demonstrating how the combination of ontology engineering and logical reasoning significantly expands the representational power of the knowledge graph.

3 Insights

3.1 SPARQL based Analysis

To analyze the Human-AI collaboration scenarios, we executed a set of SPARQL queries over the knowledge graph using GraphDB. These queries were designed to explore how autonomy, explainability, and agreement influence outcome quality.

3.1.1 Query 1: Autonomy and Explainability vs Outcome

```
SELECT ?interaction ?autonomy ?explainability ?outcome
WHERE {
  ?interaction rdf:type hi:Interaction .
  ?interaction hi:hasAutonomyLevel ?autonomy .
  ?interaction hi:hasExplainabilityLevel ?explainability .
  ?interaction hi:hasOutcomeQuality ?outcome .
}
```

	interaction ↕	autonomy ↕	explainability ↕	outcome ↕
1	hi:S1_DrHouse	hi:HighAutonomy	hi:LowExplainability	hi:BadOutcome
2	hi:S2_DreamTeam	hi:MediumAutonomy	hi:HighExplainability	hi:GoodOutcome
3	hi:S3_RobotTeaching	hi:HighAutonomy	hi:HighExplainability	hi:GoodOutcome
4	hi:S4_RobotTeaching	hi:LowAutonomy	hi:HighExplainability	hi:MixedOutcome
5	hi:S5_Research	hi:LowAutonomy	hi:LowExplainability	hi:BadOutcome
6	hi:S6_Research	hi:MediumAutonomy	hi:PartialExplainability	hi:MixedOutcome

Figure 2: Interaction configurations showing autonomy, explainability, and resulting outcomes.

This query provides an overview of how different combinations of autonomy and explainability relate to outcomes. The results show that high explainability is consistently associated with positive or improved

outcomes, while low explainability correlates with negative outcomes, regardless of autonomy level. In particular, the worst case scenarios combine low explainability with either very high or very low autonomy.

3.1.2 Graph based Visualization of Failure Cases

To complement the tabular analysis, we generated a focused graph visualization of failure scenarios using a SPARQL CONSTRUCT query. The goal of this visualization is to highlight the structural relationships between interaction instances and the factors associated with bad outcomes.

```
CONSTRUCT {
  ?interaction hi:hasAutonomyLevel ?autonomy .
  ?interaction hi:hasExplainabilityLevel ?explainability .
  ?interaction hi:hasAgreementLevel ?agreement .
  ?interaction hi:hasOutcomeQuality hi:BadOutcome .
}
WHERE {
  ?interaction rdf:type hi:Interaction .
  ?interaction hi:hasOutcomeQuality hi:BadOutcome .
  ?interaction hi:hasAutonomyLevel ?autonomy .
  ?interaction hi:hasExplainabilityLevel ?explainability .
  ?interaction hi:hasAgreementLevel ?agreement .
}
```

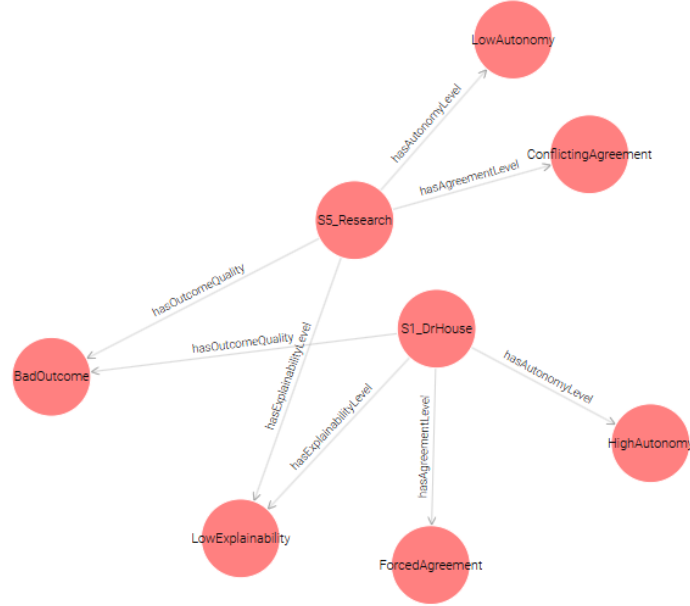


Figure 3: Graph visualization of failure scenarios showing relationships between interaction properties and bad outcomes.

The resulting graph focuses exclusively on interactions that lead to bad outcomes, providing a compact and interpretable view of failure conditions. The visualization reveals that both failure scenarios share low explainability, while differing in autonomy levels and agreement types (forced versus conflicting agreement). This highlights low explainability as a common critical factor in unsuccessful Human-AI collaboration, while also showing that different forms of misalignment between human and AI agents can lead to similar negative results.

Compared to tabular representations, this graph based view emphasizes the structural relationships within the knowledge graph and makes it easier to identify shared patterns across failure cases.

3.1.3 Query 3: Agreement vs Outcome

```
SELECT ?agreement ?outcome (COUNT(?interaction) AS ?count)
WHERE {
  ?interaction rdf:type hi:Interaction .
  ?interaction hi:hasAgreementLevel ?agreement .
  ?interaction hi:hasOutcomeQuality ?outcome .
}
GROUP BY ?agreement ?outcome
ORDER BY DESC(?count)
```

	agreement	outcome	count
1	hi:AlignedAgreement	hi:GoodOutcome	"2"^^xsd:integer
2	hi:ConflictingAgreement	hi:MixedOutcome	"2"^^xsd:integer
3	hi:ForcedAgreement	hi:BadOutcome	"1"^^xsd:integer
4	hi:ConflictingAgreement	hi:BadOutcome	"1"^^xsd:integer

Figure 4: Distribution of agreement levels and corresponding outcomes.

This aggregated query shows a clear relationship between agreement and outcome quality. Aligned agreement consistently leads to good outcomes, whereas conflicting agreement is associated with both mixed and bad outcomes. Forced agreement leads exclusively to bad outcomes. This demonstrates that alignment between human and AI agents is a critical factor for successful collaboration.

3.1.4 Query 4 : Team Composition Across Use Cases

```
SELECT ?useCase ?domain ?agentType (COUNT(DISTINCT ?member) AS ?teamSize)
WHERE {
  ?useCase rdf:type hi:UseCase ;
    hi:hasDomain ?domain ;
    hi:hasHITeam ?team .

  ?team hi:hasMember ?member .

  OPTIONAL { ?member rdf:type hi:HumanAgent .
    BIND("HumanAgent" AS ?agentType) }
  OPTIONAL { ?member rdf:type hi:ArtificialAgent .
    BIND("ArtificialAgent" AS ?agentType) }

  FILTER EXISTS {
    ?team hi:hasMember ?aiMember .
    ?aiMember rdf:type hi:ArtificialAgent .
  }
}
GROUP BY ?useCase ?domain ?agentType
ORDER BY ?useCase ?agentType
LIMIT 10
```

The results show that Human-AI teams across different use cases consistently include both human and artificial agents, reinforcing the hybrid nature of the modeled interactions and confirming that the ontology captures genuine collaborative settings. By ordering the results by use case and agent type, the query clearly presents both agent categories side by side, making it easier to compare their contributions within each team. The observed differences in team size across agent types highlight how participation varies between humans and AI systems, providing insight into the structure of

	useCase	domain	agentType	teamSize
1	hiAdaptiveCybersecurityUseCase	http://dbpedia.org/resource/Com...	'ArtificialAgent'	*1**xsd:integer
2	hiAdaptiveCybersecurityUseCase	hi:SecurityDomain	'ArtificialAgent'	*1**xsd:integer
3	hiAdaptiveCybersecurityUseCase	hi:CybersecurityExpert	'ArtificialAgent'	*1**xsd:integer
4	hiAdaptiveCybersecurityUseCase	http://dbpedia.org/resource/Com...	'HumanAgent'	*3**xsd:integer
5	hiAdaptiveCybersecurityUseCase	hi:SecurityDomain	'HumanAgent'	*3**xsd:integer
6	hiAdaptiveCybersecurityUseCase	hi:CybersecurityExpert	'HumanAgent'	*3**xsd:integer
7	hi:AlexLearningSupportUseCase	http://dbpedia.org/resource/Educ...	'ArtificialAgent'	*2**xsd:integer
8	hi:AlexLearningSupportUseCase	hi:EducationDomain	'ArtificialAgent'	*2**xsd:integer
9	hi:AlexLearningSupportUseCase	http://dbpedia.org/resource/Educ...	'HumanAgent'	*4**xsd:integer
10	hi:AlexLearningSupportUseCase	hi:EducationDomain	'HumanAgent'	*4**xsd:integer

Figure 5: Distribution of team Composition across use cases.

collaboration and enabling further analysis of how these compositions relate to interaction dynamics and outcomes.

3.1.5 Additional Queries

In addition to the core queries, further SPARQL queries were used to explore interaction patterns, identify conditions leading to successful outcomes, and compare results across different contexts (e.g., educational vs. research settings). These supplementary analyses support the robustness of the findings and provide additional qualitative insights into the dynamics of Human-AI collaboration.

3.2 KG Metrics

Due to the limited number of scenario instances, we complement the analysis by computing metrics on the full knowledge graph. After removing literals, blank nodes, and schema level constructs, the resulting graph consists of 547 nodes and 979 edges.

Graph Density and Connectivity The overall density of the graph is 0.0033, indicating a highly sparse structure. This means that only a very small fraction of all possible connections between nodes is realized. Such sparsity is expected for ontology based knowledge graphs, where entities are connected through selective semantic relationships rather than dense, arbitrary links.

The average node degree is 3.58, showing that each node is connected to only a small number of other nodes. The mean degree centrality (0.0066) further confirms that, on average, nodes are connected to less than 1% of all possible nodes. The degree distribution (Figure 6) reveals a skewed pattern, where most nodes have low degree, while a small number of nodes act as highly connected hubs.

Local Structure The average clustering coefficient is 0.080, which is significantly higher than the global density. This indicates that, although the graph is globally sparse, local neighborhoods exhibit a higher tendency to form connections. In other words, nodes that are connected to a common node are more likely to be connected among themselves, suggesting the presence of semantically coherent local structures within the graph.

Centrality Analysis To identify structurally important nodes, we computed degree based and path based centrality measures. The mean betweenness centrality is very low (2.3×10^{-5}), indicating that most nodes do not play a significant role in connecting different parts of the graph. However, the maximum betweenness value (0.00057) highlights the presence of a small number of nodes that act as important bridges between otherwise weakly connected regions.

In addition, PageRank centrality was used to identify globally influential nodes. The results (Figure 7) show that only a few nodes, such as Communication, ApplyLinearRegression, and GuidedRobotInteraction, receive significantly higher importance scores, confirming the presence of hub like structures. These nodes represent key concepts that are frequently referenced and play a central role in the organization of the knowledge graph.

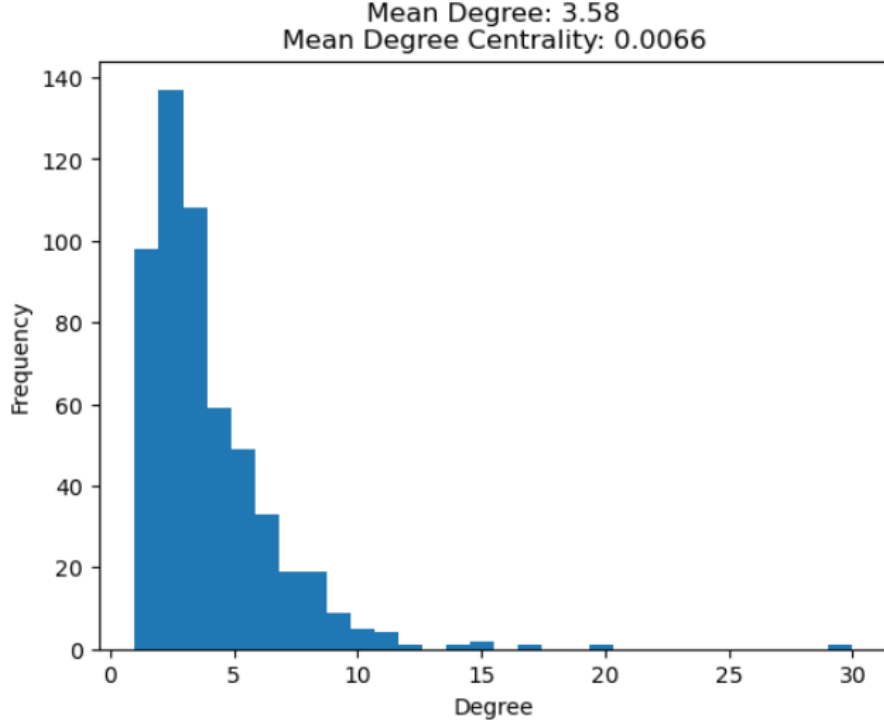


Figure 6: Degree distribution of the full knowledge graph.

Summary Overall, the full knowledge graph exhibits a sparse but structured topology, characterized by low global connectivity, moderate local clustering, and the presence of a small number of highly influential nodes. This structure is typical for semantic knowledge graphs and reflects the deliberate modeling of meaningful relationships rather than uniform connectivity.

3.3 Machine Learning for KG (Link Prediction)

To complement the symbolic analysis of the knowledge graph, we applied a sub symbolic approach using link prediction. The goal was to learn latent representations of entities and relations and evaluate the model’s ability to infer missing triples.

Data Preparation. Before training, we performed a cleaning step to reduce bias from trivial or overly frequent relations. The most frequent predicates in the graph were `rdf:type` (712 occurrences), `rdfs:label` (444), `owl:sameAs` (127), `rdfs:comment` (89), `rdfs:range` (56), and `rdfs:domain` (56). These relations primarily encode schema level or annotation information rather than meaningful relational structure. Therefore, we removed these five predicates from the training data to focus the model on more informative relationships.

Model Training. We trained a ComplEx model using the PyKEEN framework[10, 11]. Multiple configurations were evaluated by varying the number of epochs and learning rates. Figure 8 shows the training loss curve, which decreases smoothly over time, indicating successful optimization of the training objective.

Although the training loss continues to decrease with more epochs, the test MRR does not improve and slightly decreases after 200 epochs (see Figure 9). This indicates overfitting, where the model starts memorizing the training data rather than generalizing to unseen triples. Therefore, we selected the model trained for 200 epochs as the final model.

Evaluation and Baseline Interpretation. The knowledge graph contains 516 entities. In such a setting, a random baseline for Hits@10 is approximately:

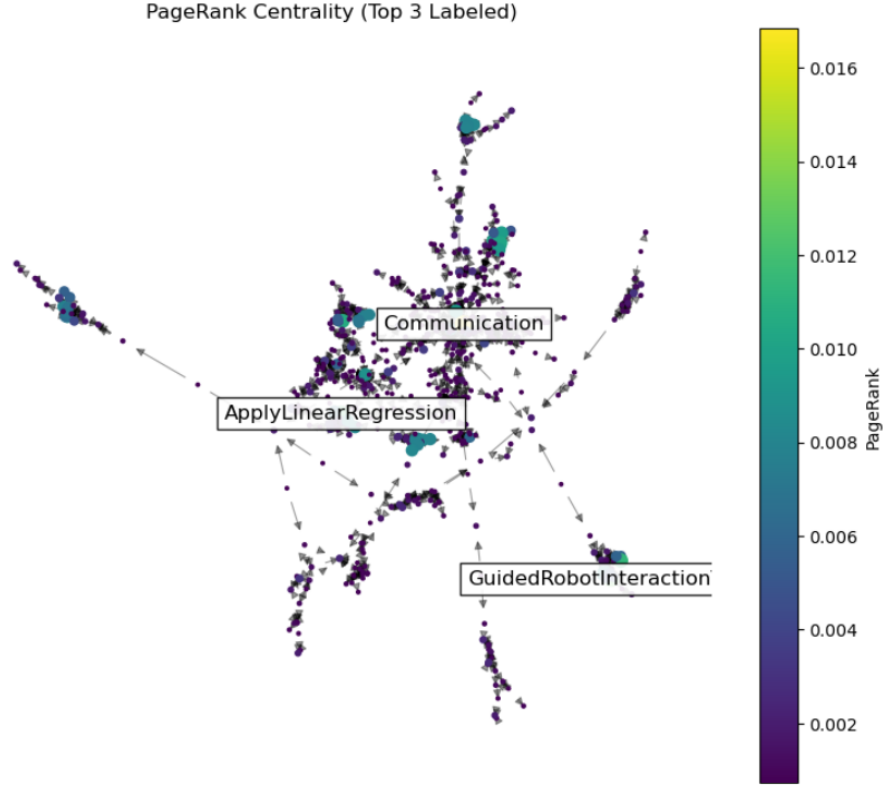


Figure 7: PageRank centrality visualization highlighting the most influential nodes.

$$\text{Hits@10}_{\text{baseline}} = \frac{10}{516} \approx 0.0194$$

Our best model achieves a Hits@10 of around 0.0278, which is only slightly above this baseline. This indicates that the model has limited generalization capability and struggles to rank correct triples significantly higher than random candidates.

This apparent contradiction—low loss but weak ranking performance—can be explained by the nature of the training objective. The model optimizes a margin based loss that separates positive and negative triples locally, but this does not necessarily translate into strong global ranking performance. In other words, the embeddings can fit the training data well while still failing to generalize effectively across the full entity space.

Frequent Relations and Prediction Setup. After training, we selected a subset of 19 triples for qualitative link prediction analysis. The selection strategy was twofold. First, we considered frequently occurring relations in the knowledge graph, such as **hasCapability** and **hasMember**, as these provide sufficient training signal and are well represented in the learned embeddings. Second, we included relations that are semantically important for our scenarios, such as **hasOutcomeQuality**, even if they are less frequent. This ensures that the evaluation covers both statistically dominant patterns in the graph and conceptually relevant aspects of the Hybrid Intelligence scenarios.

Qualitative Link Prediction Results. Table 1 shows example predictions generated by the model. We observe that most predicted probabilities are highly polarized, i.e., close to either 0 or 1.

This behavior can be attributed to the embedding model’s tendency to produce high confidence scores for patterns it has seen during training, while assigning very low scores to unfamiliar combinations. The bold triples were already present in the knowledge graph, and the model correctly assigns them near perfect scores, demonstrating strong memorization of seen facts.

However, this memorization also contributes to poor generalization. For example, in the **S2_DreamTeam** scenario, the model assigns a probability of 1.0 to multiple mutually exclusive outcomes (e.g., **GoodOutcome**,

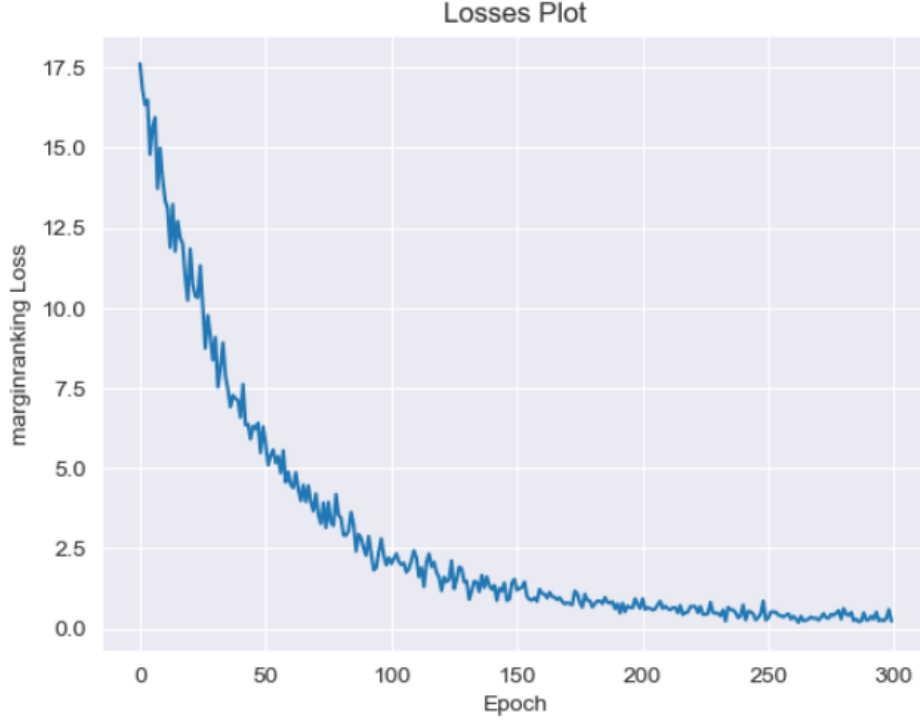


Figure 8: Training loss over epochs.

	Model	Train/Test	Epochs	Learning Rate	MRR	Hits@10	Hits@3
0	Model 1	80/20	200	0.0010	0.0194	0.0278	0.0111
1	Model 2	80/20	300	0.0010	0.0195	0.0278	0.0111
2	Model 3	80/20	200	0.0005	0.0183	0.0250	0.0139
3	Model 4	90/10	200	0.0005	0.0214	0.0278	0.0167

Figure 9: Evaluation metrics for different model configurations.

MixedOutcome, and BadOutcome).

This counterintuitive result arises because embedding based models such as ComplEx do not enforce logical constraints[10] like mutual exclusivity. Each triple is evaluated independently, and the model does not encode that certain relations should be disjoint. As a result, it can assign high scores to multiple conflicting triples if they individually resemble patterns observed in the training data.

Discussion. Overall, the results highlight key limitations of purely embedding based approaches for knowledge graphs:

- The model tends to memorize training data rather than generalize.
- Logical constraints (e.g., exclusivity of outcomes) are not captured.
- High training performance does not necessarily translate to strong ranking metrics.
- Predictions are often overconfident, leading to extreme probabilities.

These findings suggest that integrating symbolic reasoning or constraint aware methods could improve the quality and interpretability of link prediction in this setting.

The presence of OWL constraints such as disjointness, functional properties, and value restrictions further emphasizes the gap between symbolic reasoning and embedding-based predictions.

Triple	Probability
(S1_DrHouse, hasOutcomeQuality, GoodOutcome)	0.000
(S1_DrHouse, hasOutcomeQuality, MixedOutcome)	0.000
(S1_DrHouse, hasOutcomeQuality, BadOutcome)	1.000
(S2_DreamTeam, hasOutcomeQuality, GoodOutcome)	1.000
(S2_DreamTeam, hasOutcomeQuality, MixedOutcome)	1.000
(S2_DreamTeam, hasOutcomeQuality, BadOutcome)	1.000
(S6_Research, hasOutcomeQuality, GoodOutcome)	0.000
(S6_Research, hasOutcomeQuality, MixedOutcome)	1.000
(S6_Research, hasOutcomeQuality, BadOutcome)	0.919
(Human_Expert, hasCapability, Communication)	1.000
(RobotAgent, hasCapability, PedagogicalSkills)	0.000
(ChildAssessmentTask, requiresCapability, BehavioristApproach)	1.000
(BrainAgeAssessmentTask, requiresCapability, ClinicalJudgment)	0.000
(S2_DreamTeam, hasAgentInvolved, Human_Expert)	0.005
(S3_RobotTeaching, hasAgentInvolved, RobotAgent)	0.000
(GuidedRobotInteractionExecution, performedBy, RobotAgent)	0.000
(ChildAssessmentExecution, performedBy, Human_Expert)	0.000
(RobotAssistedLearningTeam, hasMember, Human_Expert)	1.000
(RobotAssistedLearningTeam, hasMember, RobotAgent)	0.414

Table 1: Example link prediction probabilities. Bold entries correspond to triples already present in the training data.

4 Research Proposal

4.1 Problem Definition

The link prediction experiments presented in Section 3.3 reveal a fundamental limitation of purely embedding based approaches when applied to ontology grounded knowledge graphs. The ComplEx model assigns high probability scores to logically mutually exclusive outcomes simultaneously. For instance, predicting that scenario S2_DreamTeam has both Good Outcome and Bad Outcome with probability 1.0. This failure is not because of poor training, but rather a structural consequence of how neural KG embedding models are designed to operate. They optimize a scoring function over individual triples in isolation and have no mechanism for enforcing semantic constraints such as disjoint axioms, functional properties, or cardinality restrictions. In a KG modeling Human-AI collaboration—where each interaction has exactly one outcome quality and where HighAutonomy and LowAutonomy are mutually exclusive, this renders the model’s predictions unreliable for downstream reasoning or decision support. The core problem is how can link prediction over HI collaboration knowledge graphs be performed in a way that respects the semantic constraints encoded in the underlying ontology?

4.2 Motivation

This problem matters well beyond the immediate experiments. As Human-AI collaboration systems are deployed in high stakes domains, there is growing interest in using knowledge graphs to audit, explain, and improve collaborative workflows. Link prediction over such graphs could support several practical tasks like forecasting outcomes for new HI configurations before deployment, recommending interaction designs by identifying which combinations of autonomy and explainability tend to produce aligned outcomes, and detecting inconsistencies in how collaboration scenarios are documented.

None of these applications are viable if the predictive model systematically violates the ontological constraints that define what a valid scenario looks like. Constraint aware KG completion is therefore not a refinement, it is a basic requirement for any HI analysis that aims to be trustworthy.

4.3 Related Work

KG embedding models such as TransE, DistMult, and ComplEx perform well on standard benchmarks[10] like FB15k and WN18RR, but these benchmarks place few logical constraints on what valid predictions look like. The challenge of making neural embeddings respect symbolic ontological rules has been

recognized as a genuinely difficult problem in its own right. Hohenecker and Lukasiewicz articulate the core tension, statistical learning and logical reasoning each capture something the other misses [12], and neither is sufficient alone. Several neuro symbolic frameworks have been proposed to bridge this gap. KGRL incorporates OWL class constraints into the training process as soft penalties, while Neural LP learns logical rules in a differentiable form that can be grounded directly in the knowledge graph. More recent contributions include DRUM, which learns rules end to end and composes naturally with embedding models, and BeliefKG, which combines probabilistic soft logic with neural representations to enforce relational constraints at inference time.

Work on post hoc explanation of KG embeddings has further highlighted that even when models produce correct predictions, the implicit reasoning paths they rely on may violate ontological commitments. Tiddi and Schlobach provide a comprehensive survey of KG based approaches to AI explanation [14], establishing the broader context in which constraint aware completion becomes relevant to HI research.

Recent research has moved beyond traditional translational or bilinear models toward ontology-aware embeddings, which incorporate the ontology schema (TBox) into the embedding space. Methods such as OWL2Vec and RDF2Vec capture the ontology’s semantics by leveraging graph structure and logical axioms during representation learning [13]. More broadly, hybrid approaches combining symbolic knowledge with machine learning have been proposed to improve reasoning and interpretability in knowledge-driven AI systems [9, 8]. However, these approaches often struggle with the dynamic nature of Human–AI interactions, where qualitative values (e.g., “High Autonomy”) act as functional constraints rather than simple labels. Our work addresses this by focusing on disjointness preserving embeddings, ensuring that concepts explicitly defined as separate in the ontology cannot be incorrectly merged in the embedding space.

4.4 Research Hypothesis and Questions

We hypothesize that integrating ontological constraints specifically disjoint axioms, functional property declarations, and cardinality restrictions encoded in the HI ontology, into the training objective or inference procedure of a KG embedding model will improve the logical consistency of link predictions without sacrificing accuracy on unconstrained triples. This gives rise to three research questions:-

1. RQ1: How often do standard knowledge graph models break logical rules when predicting new links in our health intervention knowledge graph?
2. RQ2: Can we reduce these logical errors during training by penalizing rule violations, without hurting overall prediction quality?
3. RQ3: Does training with these rules help the model better predict important outcome related information, particularly those involving outcome quality?

4.5 Proposed Approach

We propose a constraint aware ComplEx training procedure. Beginning from the HI knowledge graph described in Section 2, the approach proceeds in three steps:-

1. Extract the rules from the ontology using automated queries. We pull out three types of rules, Disjointness rules, Functional properties and Maximum cardinality limits.
2. Turn rules into math penalties that work with the model’s internal representations (embeddings).
3. Add these penalties to the training process alongside the normal learning objective.

In simple terms, the model learns using this combined formula:- $\text{Total Loss} = \text{Normal Training Loss} + (\text{Penalty Weight} \times \text{Rule Breaking Penalty})$

The rule breaking penalty increases when the model tries to:-

- Give high scores to things that shouldn’t go together (like entities from incompatible classes).
- Predict multiple values for something that should only have one value.

Because constraint extraction is implemented as a SPARQL query over the ontology, the approach generalizes automatically to any OWL annotated KG without manual constraint specification.

We also built a simpler backup method: let the model make predictions normally, then check each prediction against the ontology using a reasoning tool (HermiT). If a prediction breaks a rule, throw it out and use the next best option instead. This checking method works with any model and gives us a good comparison point for evaluating whether training with constraints is actually better than just filtering predictions afterward.

4.6 Proposed Evaluation

We'll evaluate our approach using two measures, how accurate the predictions are and how logically correct they are.

4.6.1 Prediction Accuracy

We'll measure three standard metrics (MRR, Hits@3, and Hits@10) on a test set containing 20% of the data. We'll make sure the test set includes a good mix of different relationship types, especially outcome quality relationships. We'll compare four different methods:-

- Standard ComplEx. The basic model with no constraints (our baseline).
- ComplEx with constraint training. Our main approach that learns to avoid rule violations.
- ComplEx with post filtering. Standard model but we check and remove bad predictions afterward.
- Pure rule based predictions. Using only SPARQL queries (should be perfectly logical but probably less accurate).

4.6.2 Logical Correctness

We'll calculate a Constraint Violation Rate (CVR): out of the top-10 predictions for each test case, what percentage break at least one logical rule? We'll verify rule violations using the HermiT reasoning tool. Success means achieving a much lower CVR than the baseline while keeping MRR scores competitive.

To ensure the findings are not overfitted to our specific use cases, we will augment the evaluation by testing the constraint-aware ComplEx model on a subset of the DB100K or FB15k-237 datasets. We will manually inject synthetic functional and disjointness constraints into these benchmarks to measure if our "Rule Breaking Penalty" consistently reduces the Constraint Violation Rate (CVR) across different graph topologies and scales.

5 Reproducibility

To ensure the reproducibility of this study, all project artifacts and implementation details are documented and made available.

The extended HI-Ontology v2 is provided in Turtle format (Final.ttl), containing both the TBox definitions and the ABox instances representing the 21 extracted use cases and 6 detailed interaction scenarios. The ontology was developed using Protégé and evaluated with OWL reasoning tool HermiT [20].

The extraction of use cases followed the structured manual annotation protocol described in Section 2.2. All SPARQL queries used for analysis were executed using GraphDB 10.x, which served as the primary knowledge graph management system.

For the machine learning experiments, knowledge graph embedding models were implemented using the PyKEEN framework with a fixed random seed to ensure reproducible training behavior. Model training configurations, evaluation scripts, and data preprocessing steps are documented within the project repository.

The complete dataset, ontology files, and implementation code are publicly available in the project repository:

<https://github.com/PriyanshiDhillon/KnowledgeGraph>

6 Conclusions

This paper analyzed Human-AI collaboration using a Knowledge Graph approach, focusing on how autonomy level, explainability level, and agreement level affect outcome quality across six real world scenarios. We made three main contributions:

- Extended the HI Ontology - We added new classes and properties to version 2 of the HI Ontology, building on established standards (PROV-O, FOAF, Schema.org) and connecting our concepts to DBpedia for broader integration.
- Found key patterns through data analysis - Using structured queries, we discovered that high explainability is a better predictor of good outcomes than high autonomy alone, and that forcing agreement always leads to bad outcomes, no matter what other factors are present.
- Tested machine learning predictions - We applied a ComplEx model to predict missing links in the graph. While the model learned well during training, it performed barely better than random guessing on the test set and frequently violated logical rules—for example, predicting that an outcome could be both "good" and "bad" at the same time. This problem directly motivates our proposal in Section 4 for constraint aware training.

Our analysis of the knowledge graph structure showed it's sparse but well organized [13], the density is 0.0033 with a clustering coefficient of 0.080, meaning concepts form tight, meaningful clusters around central topics like Communication, ApplyLinearRegression, and GuidedRobotInteraction. These hubs reflect what the HI research literature emphasizes most.

6.0.1 Limitations

Several important limitations should be noted:

- Small sample size - With only six scenarios, we can't draw statistically valid conclusions. Our SPARQL findings should be seen as structured observations, not generalizable facts.
- Limited training data - The link prediction model suffers from the small graph size, and the ComplEx architecture we used doesn't naturally respect the logical rules (like mutual exclusivity) that are encoded in the ontology.
- Limited external coverage - Although the ontology now includes links to both DBpedia and Wikidata, the number of external connections is still relatively small compared to large-scale knowledge graphs. This limits opportunities for extensive federated querying and integration with broader semantic resources.

6.0.2 Future Work

We plan to address these limitations in three phases:

- **Medium term:** Implement and test the constraint aware training method proposed in Section 4. This would show whether adding logical rules during training actually improves consistency while maintaining prediction quality.
- **Long term:** Extend the ontology to capture how things change over time (like how agreement and trust develop across multiple interactions) and add richer models of agents that include expertise, communication styles, and cognitive load. This would move us toward a knowledge graph that could support real time decision making in actual Human-AI collaboration systems.

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